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Developing a Computer Model for Optimization of Drilling and Blasting In Open Pit Mines from the Viewpoint of Technical, And Environmental Parameters

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Drilling and blasting are among the most significant unit operations in open pit mining projects. The main objective of utilizing explosives in mines is to fragment and move the rock mass (ore or waste) with the lowest possible cost. Therefore always a certain degree of fragmentation is demanded from the blasting engineer. On the other hand, there are often typical safety considerations in many surface mines that must be taken into account in blasting design procedure. Ground vibration, Flyrock (rock fragment projection) and Airblast (air overpressure) are the most critical environmental side effects of mine blasting which can potentially inflict severe hazards and great losses. Numerous research works have been conducted in various aspects of mine blasting technology such as conventional blast design, fragmentation prediction and environmental side effects control methods; and many theories have already been put forward. Nevertheless, a comprehensive model for blasting design with the aim of attaining optimum fragmentation and effective containment of environmental effects simultaneously is still missing. In this paper, a new approach to open-pit mining blast design is introduced; and a new software code, named BlastDesign, is presented with the purpose of assisting D&B engineers in choosing optimum drilling diameter and designing optimal pattern for blast operation in new mines and quarries and other projects involving bench blasting. The software recommendations are based on incorporation and integration of existing distinct models and relationships in various aspects of blasting with the aim of achieving optimum fragmentation and fulfilling strict safety and environmental limitations concurrently.

Introduction

Drilling and blasting are two of the most important and conspicuous unit operations in open pit mining. The rock body, whether ore or waste shall be initially broken into pieces and swelled and moved in order to become transportable to the final destination of mineral processing plant or waste dump. On other hand, ore minerals are suitable for the subsequent mineral processing operations (comminution, etc.) only when they are fragmented in apt sizes through the mining step. Therefore, virtually in every open pit mine, the engineer in charge of blasting is supposed to be able to predict and control the degree of fragmentation of the blasted rock beforehand, and also able to make necessary modifications in it. On the other hand each and every blast operation in mining or civil projects entails various environmental effects and safety considerations which have to be thoroughly considered in advance. Ground vibration, air overpressure (technically known as airblast among miners), and fragmented rock projection (known as flyrock) are among the most major, conspicuous environmental effects of blasting operations in open pit mines which can pose serious safety risks for nearby facilities, personnel and equipment. In this research, a new software code named BlastDesign is presented which introduces a new approach in open pit blasting design and optimization with regard to attaining desired rock fragmentation and also restricting blast adverse environmental effects and accounting for safety parameters [1].

At this initial step the BlastDesign software is introduced. This software is developed based on the results of authors' research with the aim of assisting blast engineer in order to design and optimize the open pit blasting pattern more accurately and efficiently with regard to the required degree of fragmentation and allowable limits of blast safety/environmental impacts (ground vibrations, air overpressure, and rock projection). In Figure 1, the main window of BlastDesign software is depicted. It enables the user to select the results or calculation mode they demand for. There are several buttons representing distinct choices according to different objectives a blasting engineer might need during the blast design and optimization procedure. The options are as follows, from the top to the bottom:

- Conventional Blast Design
- Blast Design Control
- Blast Design according to the Required Fragmentation (a technical parameter)
- Blast Design according to the allowable limits for the Blast-induced Ground Vibration
- Blast Design according to the allowable limits for the Blast-induced Air Overpressure (airblast)
- Blast Design according to the allowable limits for the Blast-induced Rock Projection
- Comprehensive Model taking all technical/safety parameters into account



Figure 1: Initial primary window of the program

Conventional Blast Design with Pre-determined Hole Diameter

The first module of the software deals with conventional blast design in active open pit mines with pre-determined blasthole diameter. In figure 2, one can see the input data and results of this module of the software for conventional blast design. This is the most widely-used method in bench blasting when the drilling diameter is already known and applied in the mine. This is the case in most active open pit mines where a drilling wagon or drilling rig with known specifications is in place; so the blasting engineer has to adopt his/her own requirements with this machine. Burden (the distance from a single row of blast-holes to the face of the excavation) is the foremost and most important parameter that must be worked out in bench blasting design. There are numerous relationships put forward by different researchers in order to design blast blasting pattern according to a predefined drilling diameter.

BlastDesign Software makes use of seven of the most trusted and recognized methods among these relationships [2, 3]. The methods applied in BlastDesign are as follows:

- Ash design guidelines
- Bhandari relationship
- Lopez Jimeno
- Konya (1972)
- Konya (1983)
- Roustan
- Tatiya

One can consult with the references in this field in order to acquaint themselves with each of these blast design relationships. After a logical size is determined for burden, normally other parameters in open pit blasting are considered in direct relationship with the burden value; and hence that are calculated according to it with some coefficients defined for this purpose. The coefficients are, namely:

K_B : Burden to hole diameter ratio (after Ash), which is decided with regard to the rock compressive strength and density as well as explosive relative weight strength with ANFO as a reference. The recommended range for K_B is 20 to 40. For the hardest rock with the lowest strength explosive (such as ANFO), recommended value for K_B is 20; and for the soft least dense rock with the most potent explosive, the recommended value is 40.

K_S : Spacing to burden ratio ($1 \leq K_S \leq 2$)

K_H : Bench height to burden ratio which represents bench stiffness ($1.5 \leq K_H \leq 4$)

K_J : Subdrilling (over-drilling) to burden ratio; subdrilling is often necessary in order to be able to remove the bench bottom part, known as toe. ($0 \leq K_J \leq 0.5$)

K_T : Stemming to burden ratio; stemming is essential in order to confine the energy of blast inside the rock body and effectively fragment and swell the rock. The minimum recommended value for this ratio is 0.7. ($0.5 \leq K_T \leq 1.3$) [2, 3, 4]

Since none of the proposed values by different methods shall be considered as the final and perfect results, the software also calculates the mean and standard deviation of the results of different methods and also recommends a suggested value based on the mean while removing the outliers according to the confidence level preferred by the user.

Figure 2: Conventional blast design window (based on predetermined blasthole diameter)

Blast Design Control Module

The next available facility in BlastDesign software is a module to predict current or proposed blast pattern for the mine with regard to projected fragmentation and the intensity of blast environmental effects, e.g. ground vibrations, air overpressure (airblast), and maximum rock throw (flyrock). In the second window of the program shown in figure 3, user is able to predict and control the results of a proposed blasting pattern even before conducting a single test-shot in the mine environment. The software is capable of predicting the results of different drilling diameters and blasting patterns with regard to mean fragmentation (x_{50}) and safety/environmental concerns i.e. maximum possible rock projection distance (flyrock), airblast and ground vibration at a specified distance to the center of the blasting area. One shall take notice that in order to predict the ground vibration at a specified point, the software would demand for the site-specific vibration propagation constants (K and β) that are experimental constants in the vibration attenuation equation and must be empirically decided for the area under study beforehand [3]. There are also two different approaches for determining the level of air overpressure (airblast) in the mine area: one according to the relationship put forward by W.Hustrulid [3], and the other according to suggested formulae by S.Bahandari [5]. The software also predicts the level of fragmentation given a specified drilling diameter and blasting pattern according to the rock type and explosive relative strength. The control module makes use of the Kuz-Ram fragmentation prediction model [2] to calculate the resultant mean fragmentation.

Blast Design Control

Control Parameters

- ☒ Calculate fragmentation
- ☒ Calculate ground vibration
- ☒ Calculate airblast
- ☒ Calculate flyrock

Rock strength...

Rock Factor (A)

Site-specific vibration propagation constants:

K: Beta:

Vibration distance m

Airblast estimation method:

☒ Bhandari derived approach

☐ Hustrulid derived approach

Airblast distance m

Flyrock estimation method:

☒ Ideal bench blasting

☐ Crater blasting

Number of simultaneous blast-holes

Explosive

Select explosive type:

Explosive Density gr/cm³

Relative weight strength of explosive

Blasting Parameters

Drilling diameter mm

Drilling angle

Burden m

Spacing m

Bench height m

Subdrilling m

Stemming m

Results

Flyrock maximum throw distance m

Peak Particle Velocity mm/s

Airblast intensity dB

Mean fragmentation (x50) cm

Results

Figure 3: Blast pattern control with regard to fragmentation and environmental effects

Fragmentation-wise Blast Design

In the next module, the software enables the user to design a blast pattern according to the desired degree of fragmentation which is supposed to be attained by bench blasting. The module is called "Optimum Fragmentation Blast Design" and is shown in Figure 4; the user shall input some general specifications of the rock and explosive: namely UCS and rock factor (based on Kuz-Ram model) for the rock type, density and relative weight strength (RWS) for the explosive. The software also demands the user to specify the drilling inclination and pattern (staggered or rectangular) as well as the chosen ratios between blasting parameters (K_B , K_H , K_S , K_J , and K_T) according to their engineering judgement and wishes. As the ultimate input parameter, the user will also stipulate the degree of fragmentation they would require from the blast operation in open pit, whether in the form of 50% passing size (x_{50}), or any other specification (x_{80} or else). Finally the software provides the user with the optimum drilling diameter and blasting pattern with the purpose of attaining the user-specified fragmentation. The complete blast design data includes burden, spacing, bench height, subdrilling, stemming, and the weight of explosive in each hole. There are also calculations for the specific charge and specific drilling for the whole drilling pattern in the results section.

Optimum Fragmentation Blast Design

Rock and Explosive

Rock compressive strength MPa

Rock factor (A) ☒ Precise value ☐ UCS indicated value

Select explosive type:

Explosive Density (gr/cm³)

Relative weight strength of explosive

Blasting Parameters

Drilling Pattern: ☒ Staggered ☐ Rectangular

Drilling inclination (0 to 20 degrees)

K_B : Burden to hole diameter ratio

K_S : Spacing to burden ratio

K_H : Bench stiffness ratio

K_J : Subdrilling to burden ratio

K_T : Stemming to burden ratio

Required Fragmentation

☒ x50 cm ☐ Other size cm

Uniformity index Passing percent(%)

Results

Drilling diameter mm

Burden m

Spacing m

Bench height m

Subdrilling m

Stemming m

Mass of charge in each hole kg

Specific charge gr/m³

Specific drilling mm/m³

Results

Figure 4: Blast design based upon optimum desired fragmentation

Blast Design according to Ground Vibration Limitation

The software also has capabilities in order to design and optimize the drilling and blasting pattern according to the different safety considerations and environmental limitations of the blast area. As discussed earlier, the most important of these are: ground vibration, air overpressure (airblast), and maximum rock throw (known as flyrock). The blasting design and optimization process according to each of these three different environmental concerns is defined in distinct module of the software, and also in a comprehensive integrated model as the final resolution of the software. We start with introduction to the first of these modules regarding blast design with ground vibrations in mind. One of the most conspicuous and important safety and environmental limitations in open pit mines is the intensity of the ground vibrations induced by blasting. There might be sensitive structures in the vicinity of the mine area such as the processing plant or the buildings related to the mine; or generally the ground vibrations shall be restricted due to the adverse environmental effects it could have on the wildlife of the surrounding area. The ground vibration at each point near the blasting area is dependent on the weight of charge detonated in each delay interval and the distance of that point to the single blasthole (or the center of a blasting area). It is approximated by an attenuation equation; and there are attenuation coefficients (K , and β) that shall be experimented in the mine area in order to be able to predict the blast-induced ground vibration correctly in the next shots. But there are maximum limits for these coefficients; so they are applicable for the areas with unknown (untested) attenuation coefficients and results in the most conservative outcomes with regard to the drilling diameter and consecutive blasting parameters according to the ground vibration threshold in the mine environment. One can observe the module for ground vibration cautious blast design in the software in figure 5. In addition to the explosive properties, and desired ratios between blasting parameters, the user also has to specify the number of simultaneous blast-holes in order to enable the software to calculate the weight of charge detonated in each delay interval. The software also suggests the standard threshold values for the intensity of ground vibration with regard to the distance according to OSMRE recommendations [3, 6], but the user is able to change the value according to his/her engineering judgement and the site-specific limitations.

Blasting Parameters		Results	
Explosive density (gr/cm ³)	0.85	Drilling diameter	326 mm
Drilling inclination (0 to 20 degrees)	0	Burden	9.78 m
Kb: Burden to hole diameter ratio	30	Spacing	11.25 m
Ks: Spacing to burden ratio	1.15	Bench height	25.44 m
Kh: Bench stiffness ratio	2.6	Subdrilling	2.94 m
Kq: Subdrilling to burden ratio	0.3	Stemming	6.85 m
Kt: Stemming to burden ratio	0.7	Mass of charge in each hole	1528.1 kg
Vibration Control		Specific charge	546 gr/m ³
Distance to the nearest sensitive structure	500 m	Specific drilling	10 mm/m ³
PPV safety limit	25 mm/s	Results	
Number of simultaneous blast-holes	10		
Site-specific vibration propagation constants:			
K:	205.6		
Beta:	1.508		
☐ Consider the most conservative conditions			

Figure 5: Blast design with regard to containment of induced ground vibrations

In the next module, the user is given the possibility to design and optimize the blasting pattern according to the defined airblast limitations in the blast area. Two different approaches for estimating the airblast (air overpressure) and related calculations are applied in the software, namely: Approach derived from Bhandari suggestions and approach derived from Hustrulid relationship. In general, and according to the calculations done by the software one can say that the blasting design results based on Bhandari derived approach are more conservative than the results from Hustrulid derived method. In other words, it is possible to say that Bhandari relationship for prediction of airblast intensity results in greater calculations for this safety parameter at any given point with specific distance to the blasting area [1]. For this calculation as well, the user is supposed to enter the number of simultaneously-detonated blast holes in order to enable the software to calculate the maximum allowable drilling diameter for the holes according to the airblast limits. The standard threshold for airblast is 134 decibels in general; and in case a hospital or school is located near the blasting area it is lowered to 128 decibels [3, 4].

Figure 6: Airblast cautious blast design with Bhandari approach (more conservative results)

In the next module, the user is supplied with a tool to optimally design the blast according to the limitation of maximum rock throw (flyrock) in the mine environment. The user must also specify if an ideal bench blasting is applicable in the mine, or the ultimate blast is more like a crater blasting. When the benches are geometrically regular and near-perfect bench blasting is plausible in an open pit, the maximum rock throw is dependent on both blast hole diameter and specific charge of the blast; whereas crater blasting is probable in the mine site, normally the flyrock distance could be much more and it solely depends on the diameter of the blast holes.

Figure 7: Module for Flyrock (rock projection) cautious blast design

Comprehensive Model for Blast Design

Ultimately, IKIU Open Pit Blast Design software also offers a comprehensive model for optimization of blast design with regard to desired fragmentation and containment of blast harmful environmental effects and safety hazards, namely: ground vibration, air overpressure (airblast), and rock throw (flyrock). This comprehensive model is in fact a combination and integration of all aforementioned models for blast design according to required fragmentation, blast design with regard to vibration restriction, etc. But an integration algorithm was essential in order to be able to assimilate the results of different approaches into each other and arrive at the final comprehensive model. For this purpose, an innovative algorithm was put into practice. The flowchart of this algorithm is shown in the next page (figure 9). In the following depictions, one can observe some of the sequential steps of the ultimate module of the software for comprehensive model. As it is graphically demonstrated in the flowchart, at the initial stage the properties of rock and explosive (UCS, precise or UCS-indicated rock factor, explosive type or new user-defined blasting agent) are stipulated by the user.

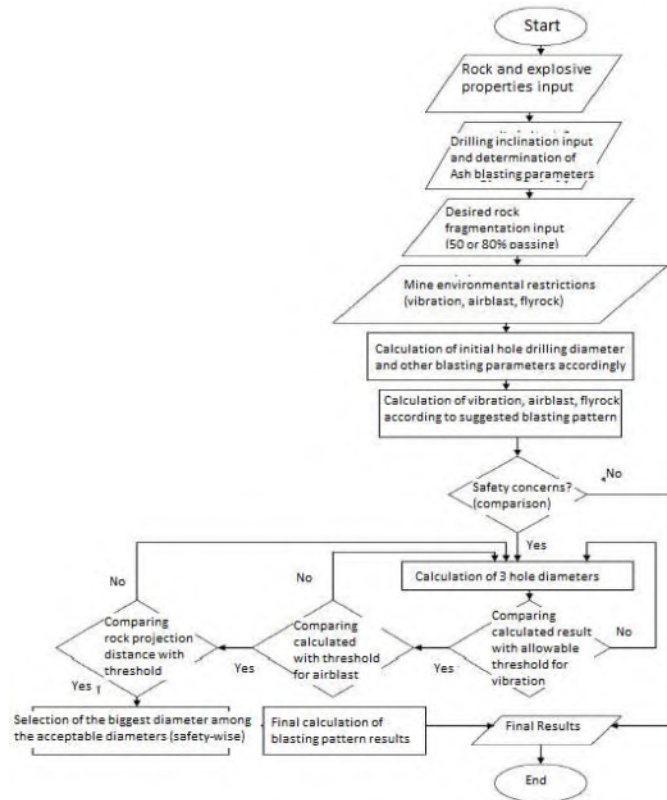


Figure 8: Flowchart of the algorithm applied in the software for comprehensive blast optimization model [1]

In the next window, the user is supposed to stipulate drilling inclination with respect to the vertical axis (0 to 20 degrees), as well as Ash proposed ratios between blasting parameters (K_B , K_H , K_J , and K_T) according to his/her engineering judgement or simply accepts the recommended values by the software which are based on rock/explosive properties and the suggestions by different researchers.

The screenshot shows the 'Drilling and Blasting' window with the following settings:

- Drilling Inclination:** Drilling angle with respect to vertical axis (degrees) is set to 5.
- Burden to Hole Diameter Ratio (K_B):** Preferred value is 30, Recommended value (with respect to UCS and explosive density) is 30.
- Bench Stiffness Ratio (K_H):** Preferred value is 2.6, Suggested value (Average value in world open pits) is 2.6.
- Spacing to Burden Ratio (K_S):** Preferred value is 1.15, Suggested value (Best Energy Distribution in Staggered Pattern (1.15cos(ϕ))) is 1.14562.
- Subdrilling to Burden Ratio (K_J):** Preferred value is 0.3, Suggested value (Medium strength rocks (50 < UCS < 150 MPa)) is .3.
- Stemming to Burden Ratio (K_T):** Preferred value is 0.7, Suggested value (Minimum recommended) is .7.

Buttons for 'Back' and 'Next' are visible at the bottom.

Figure 9: Specification of drilling angle and blasting ratios in the software

Consequently, the user must also provide the degree of fragmentation required in the mine according to the subsequent comminution steps (foremost primary crusher input size), whether in x_{50} (mean fragment size) or x_{80} (80% passing size). For conversion of x_{80} to x_{50} (applicable in fragmentation prediction models) a parameter called uniformity index is required; but the software is capable of calculating this index based on a relationship between Ash blasting ratios, provided that the deviation in drilling angle is kept negligible.

In the next window of this final module, the user can select or deselect the options for consideration of each of the safety and environmental hazards according to his/her engineering judgement regarding the risk potential of ground vibrations, airblast, and flyrock in the mining area or other bench blasting project site (see Figure 10). Regarding the ground vibration and airblast (air overpressure), the software is capable of receiving the threshold values for allowable limits and the distances to the nearest critical points. Flyrock phenomenon is also taken into account in the form of maximum rock throw distance. The software will offer the standard allowable limit for ground vibration (in the form of PPV) according the user-specified distance to the nearest critical structure. It is decided with regard to OSMRE standards. Obviously it is preferred to use precise values for the coefficients of ground vibration attenuation equation if they have been determined experimentally in the mine site earlier. But if such data are not available, the user can authorize the software to utilize the greatest possible values for these coefficients, so that the software will predict the results most conservatively with regard to the ground vibration. The user is also supposed to introduce the number of simultaneously-detonated holes according to the blasting pattern, since it is a significant factor for the prediction of blast-induced ground vibration and airblast. With regard to flyrock prediction, the user must also confirm that ideal bench blasting is plausible in the mine, and crater blasting phenomenon which results in much farther rock throw distance is not reasonably probable. If left unchecked, the software considers most conservative conditions for flyrock prediction.

Safety Considerations

Ground Vibration
 If ground vibration should be considered, tick the checkbox, and provide the needed information.
☒ Vibrations
 Distance to the nearest sensitive structure: 500 m
 PPV safety limit: 25 mm/s
☐ Consider the most conservative conditions
 Use this option, only if site-specific vibration attenuation constants are not available.
 K: 205.6
 Beta: 1.508

Airblast
 If airblast effect should be considered, tick the checkbox, and provide the needed information.
☒ Airblast
 Distance to the nearest sensitive structure: 500 m
 Airblast safety limit: 134 dB

Flyrock
 If flyrock should be considered, tick the checkbox, and enter the needed information.
☒ Flyrock
 Maximum rock throw distance: 500 m
☒ Ideal bench blast, crater blasting effect is not probable.

Simultaneous Detonations
 Number of simultaneous blast-holes: 10

Back Results

Figure 10: Final data input for the module of comprehensive model

Finally the results of the comprehensive model is demonstrated on the final window of the module. As the reader can check with the flowchart (Figure 9), the module first tries to put forward an optimum drilling diameter and resultant blasting pattern based on desired fragmentation of the user. But such results are shown as final recommendation only if the safety and environmental limitations of the mine site are met by this blasting pattern according to the safety conditions which are introduced by the user in the initial stage. If the resultant blasting pattern from desired fragmentation approach does not possess any predicted adverse consequences on the safety and environmental limitations of the mine, it will be suggested as the final recommendation of the software. Otherwise, several other blasting patterns (blasthole diameter, burden, spacing etc.) are calculated according to each environmental consideration in the mine (limitations for ground vibration, air overpressure, and flyrock); and the predicted intensities for each of the environmental effects are computed and compared with each other. Each of the blasting patterns (including blasthole diameter, burden, spacing, subdrilling, etc.) that involves no safety or environmental concerns according to the limitations set by the user will be presented finally as the ultimate results for recommended blasthole diameter and blasting pattern in the mine. In order to enable to user to compare the technical and environmental results of such pattern to what he or she had in mind beforehand, the predicted

fragmentation and environmental effects of the proposed pattern are also shown on the results window of comprehensive model. If there was a predicted safety or environmental concern (namely: flyrock, vibration, and airblast) over attaining the desired fragmentation the user had specified as input, he or she would be notified about the limiting factor in blast design; whether it is the ground vibration, air overpressure or maximum rock throw distance.

Blasting Parameters		Safety and Fragmentation	
Drilling diameter	123 mm	No Safety Problem	Vibration level 4 mm/s
Burden	3.69 m		Airblast level 124.7 dB
Spacing	4.23 m		Maximum rock throw 395.2 m
Subdrilling	1.11 m		
Stemming	2.58 m		
Bench height	9.6 m		
Mass of charge in each hole	115.5 kg		
Specific charge	770 gr/m ³		
Specific drilling	72 mm/m ³		Final x50 28 cm

Figure 11: The results window of the final comprehensive model (predicted no safety problems)

Let's see the situation in another example which is approximately the conditions in Gole-E-Gohar Open Pit Mine in south-eastern Iran and is considered as a large-scale iron ore mining project. In this mine, x_{50} (50% passing size) of 50 cm for ore is ideal for the subsequent comminution process. The vibration attenuation has been done in the mine area, and the applicable coefficients of the attenuation equation (K and β) have been determined 205.6 and 1.508 respectively [1]. Supposedly up to 10 blast-holes might be simultaneous detonated in the mine site; and mining machinery are normally stationed no closer than 600 meters to the blast face (minimum). On the other hand, though at the present stage there is no sensitive buildings prone to destructive ground vibration or airblast in the close vicinity of the blast area, for the sake of research we consider that there would be a structure susceptible to the adverse effects of ground vibration at 200-meter distance of the mine, and another structure vulnerable to airblast waves at 300-meter distance. Standard OSMRE limiting thresholds of 25 mm/s and 134 dB are also supposed for these structures respectively for ground vibration, and airblast intensity. Ratios between blasting parameters namely: K_B , K_S , K_J , and K_T are chosen according to the recommendations based on rock and explosive conditions and are set to 25, 1.15, 0.3, and 1 respectively. As the bench height in this mine is set to 15 meters from the start of the operation and it is not reasonable to alter it in this stage, a higher value of 3.42 is chosen for the ratio of bench height to burden (K_H : bench stiffness ratio) so that the final bench height remains constant at 15 m. As the results window, shown in figure 15, demonstrates in this case there would several safety concerns in the blast operation with regard to attaining the desired fragmentation of $x_{50} = 50$ cm. In fact, it is not possible to obtain this fragmentation level without compromising the supposed environmental restrictions of the mine. The software notifies the user about such conflicts also finds the most limiting factor, whether it is the ground vibration, airblast or flyrock throw distance. Furthermore, it automatically lowers the dimensions of the blasting pattern (blasthole diameter, burden, etc.) in order to fulfill the safety and environmental conditions of the blast site. It presents this new blasting pattern as the final solution of the comprehensive model for this mine, and also predicts the new fragmentation and the level of blast environmental effects. It is obvious that for this example, the level of ground vibration has been the most urgent limiting factor; and the new blasting pattern proposed by the software results in just the maximum allowable limit for this parameter (25 mm/s). But the other environmental effects would be well below the introduced thresholds of danger. In fact, the software denotes the limiting factor in yellow color.

Blasting Parameters		Safety and Fragmentation	
Drilling diameter	175 mm	Vibration level	25 mm/s
Burden	4.38 m	Airblast level	132.7 dB
Spacing	5.04 m	Maximum rock throw	531.6 m
Subdrilling	1.31 m	Note: Due to safety problems, Attaining specified fragmentation is NOT possible.	
Stemming	4.38 m	Intense Vibration Danger!	
Bench height	15 m	Intense Airblast Danger!	
Mass of charge in each hole	244.6 kg	Flyrock Danger!	
Specific charge	739 gr/m ³	Final x50 51 cm	
Specific drilling	49 mm/m ³		

Figure 12: The results window of the comprehensive model for Gole-E-Gohar example (predicted safety problems)

Results and Discussion

In this paper, a new software for optimization of drilling and blasting in open pit mines and other bench blasting projects was introduced. The software BlastDesign is capable of presenting an optimum blasting pattern for the mine (drilling diameter, burden, spacing, etc.) according to the desired degree of fragmentation in the mine and also with regard to the safety and environmental considerations of the mine site. Three of the most important safety and environmental factors regarding open pit mine blasting are ground vibrations, air overpressure (airblast), and rock fragments throw (flyrock); all of these are taken into account in the software. Therefore, the user is able to optimally design the blasting pattern according to the required degree of containment for each of them in distinct modules. But the final solution, there is also an ultimate comprehensive model which encompasses the different approaches and integrates the results of all of them, in order to enable the user to design a blast pattern for the mine in line with the desired fragmentation degree and also accounting for all major safety and environmental considerations in the mine area. As a result, it is possible to make sure that the blasting operation with the aim of attaining a specified fragmentation would not adversely affect the adjacent structures of the mine, with regard to allowable ground vibration, airblast, and flyrock. The software would automatically reduce the dimensions of the blasting pattern if such hazard is predicted according to the input safety considerations. As an example, it was found that it is not possible to arrive at desired results fragmentation-wise in Gole-E-Gohar Iron Ore Mine with supposed environmental limitations for the nearby structures of the mine.

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